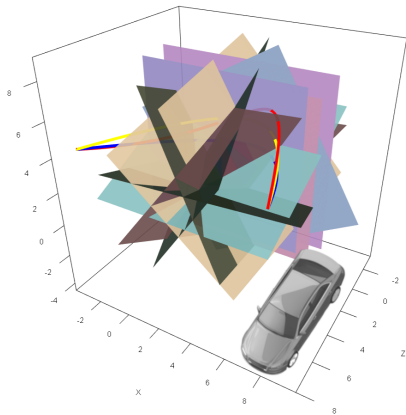


Cost Estimating: Decision-Making through Time Series Analysis and Mixed Integer Programming Optimisation

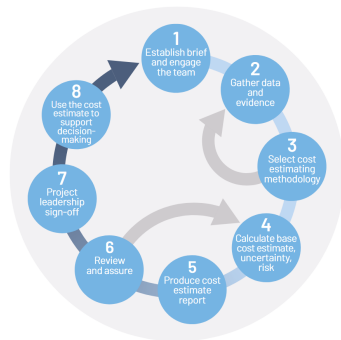
Submitted as part of the **Estimating Capability Development Scheme**



Esteban Rodriguez Yepes
29 May 2025

Objective: estimate the cost of buying a car

- ▶ To support my application for the **Estimating Capability Development Scheme (ECDS)**, a simplified **cost estimating** method was produced
- ▶ An **8-step process** for cost estimating, from government methodology, (Infrastructure and Projects Authority) served as project basis
- ▶ A **situational example** has been devised, to explain how the method would work in a real situation
- ▶ R code and spreadsheets used to produce this report can be found in my [GitHub repository](#)



The 8-Step Process for cost estimating, IPA

Project brief

A team has decided that they need to **purchase a car**, and we have been given the task to **produce a comprehensive estimate** of the final cost of purchasing the car. A few points to consider in our **project plan**:

- ▶ We must consider the **limits** of our **budget**, so a thorough look at our **funding** must be done, including the need of **financial options**
- ▶ We want a car with **low mileage**, **affordable**, and **known to be durable**, so **thorough research** on our vehicle must be done
- ▶ This purchase involves **geographical** and **transportation** factors, which incur the need of **logistics** and **insurance** (thus further costs)
- ▶ We must consider a method to make **educated choices**, considering a range of choices for cars, transportation, and finance options
- ▶ There needs to be a **version control** arrangement between all colleagues partaking in this project, for ease of work management

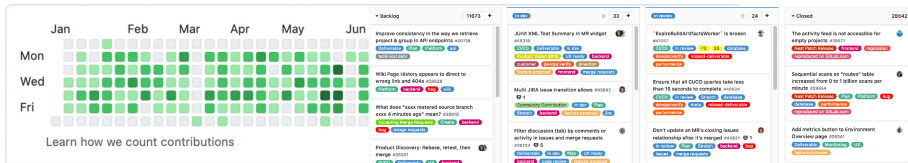
General assumptions

Our brief is summarised with a list of points and assumptions:

- ▶ The maximum length of this project is **60 months** (5 years)
- ▶ A cash flow of **£500 per month** (a net total of £30,000 for project-related costs) is guaranteed, but limited to the monthly amount
- ▶ A **business-style** car is requested, either **brand new** or **not exceeding 100,000 miles** of age. Further specifics are not requested
- ▶ We must enter a **contract** to amortise costs, in the form of either:
 1. a loan from one of **three trusted banks** we are customers of, OR
 2. a **Hire Purchase (HP)** or **Personal Contract Purchase (PCP)** from one of **two credit brokers**
- ▶ We can assume we are eligible for any of these financial options, and that funds can be issued within **5 working days**
- ▶ We can purchase the car from **any car trader** within Great Britain, but we must account for transportation costs in our estimate

Project scheduling and version control

- ▶ The areas of expertise are laid out: **data collection**, **financial analysis**, **procurement/logistics**, and **statistical programming**
- ▶ A **task schedule** on a shared platform (e.g. Teams) should be followed, with **weekly meetings** to discuss issues, progress and change
- ▶ The team should research, gather, process, and compile data and results using a shared **version control** platform, such as GitLab
- ▶ Every contributor should upload and update their work on their own project branch. If two or more individuals work on the same file, they can also **review**, **correct**, and **merge** each other's work easily



Contributions board, GitLab

Issues board, GitLab

The plan

Our plan to produce the estimate is as follows:

- ▶ We will **collect large numbers of data** from multiple car trader website listings:
 1. data on car make, model, variant, year, mileage, price, etc.
 2. data will be compiled in an Excel spreadsheet
 3. data should be **constantly checked** for and **updated**

- ▶ We will **process the data collected**, to:
 1. narrow down a **ranking** of cars with desired characteristics
 2. produce car **quality analysis** and statistics (and in particular an **estimate of price distributions**)
 3. determine a ranking of car models with best **price-mileage ratio** (akin to depreciation)

The plan, continuation

- ▶ We will make use of the **estimate of price distributions** of a car model with best age-price ratio to:
 1. ensure we choose a car that is likely to be sold at a **low price and mileage**, and has a **wide mileage distribution** (indicating longevity)
 2. ensure that the **maximum possible monthly price** of that best car model is **significantly lower** than the monthly funding limit
 3. work out the **monthly repayments**, the total repayment and the interest cost for all finance options
- ▶ We will use the produced **financial data** for the vehicle model, to:
 1. find all possible car listings for this car, and consider their **proximity**
 2. determine choices for **transportation costs** for our purchase
 3. perform **mathematical optimisation** to determine the best aggregate options of all costs

Car data intelligence

- ▶ We gather data, from **various trader listings**, with size and scope that is assumed enough for analysis. This is all compiled in **Excel**
- ▶ The most **relevant specs** gathered are: make, model, variant, car price, car badge, car seller, seller rating, seller location, year of registration, body type, mileage, engine size, engine power, transmission type, fuel type, number of owners, ULEZ-compliance, fully serviced

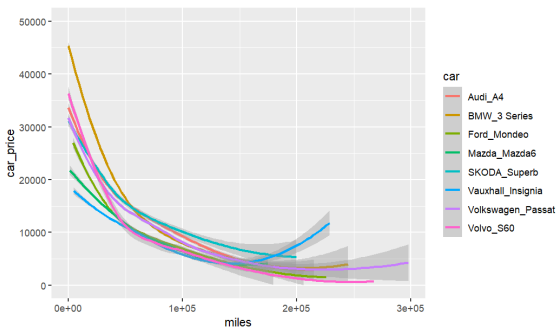
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1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	make	model	variant	car_price	car_badge	car_seller	car_seller	car_seller	year	body_type	miles	engine_vol	engine_size	engine_si	transmissi	fuel_type	num_owners	ulez	full_serv
228405	Ford	Mondeo	Titanium	11989	'Higher pri	Sascon C		4.5	reading	2014	estate	72000		2	163 ps	automatic	diesel		0
228406	Ford	Mondeo	Titanium	11995	'Good pric	Voy Trade		4.9	oldham	2014	hatchback	45234		2	163 ps	automatic	diesel	3	0
228407	Ford	Mondeo	Titanium	13059	'Good pric	Stebblings		4.4	kings lynn	2013	estate	42595		2.2	200 ps	automatic	diesel		0
228408	Ford	Mondeo	V6	2499	"	Regis Car		4.8	dunstable	2000	hatchback	103000		2.5	202 bhp	manual	petrol		0
228409	Ford	Mondeo	V6	2499		Private seller			ballymenz	1998	estate	140000		2.5	168 bhp	manual	petrol	4	0
228410	Ford	Mondeo	V6	4000	'Cat N', "	Hilltop Mc		5	north wal	2000	hatchback	89019		2.5	202 bhp	manual	petrol	4	0
228411	Ford	Mondeo	Vignale	8850	'Finance a	Private seller			lilsburn	2015	estate	119000		2	180 ps	manual	diesel	4	1
228412	Ford	Mondeo	Vignale	9990	'Good pric	Cars Direc		4.7	peterboro	2015	saloon	106000		2	180 ps	manual	diesel	2	1
228413	Ford	Mondeo	Vignale	11200	'Good pric	V12 Sports		4.4	hinckley	2017	hatchback	108150		2	180 ps	manual	diesel	2	1

Extract from a very large car sample data set, Kaggle/Guanhao Peng

First narrowing down

- ▶ We import our Excel data into **RStudio**, and to start analysis, we filter our data for **saloon** style cars, common for business vehicles
- ▶ To foresee better **repairability**, we consider eight well-known models in the UK: BMW 3 Series, Audi A4, Vauxhall Insignia, Mazda 6, Ford Mondeo, Volkswagen Passat, Volvo S60, and Skoda Superb

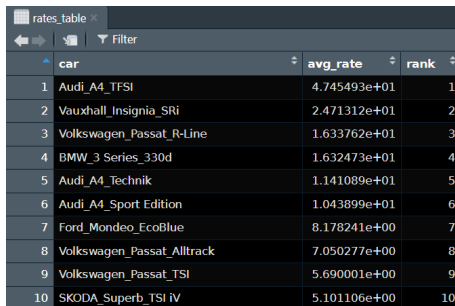
We analyse the overall **sale price-mileage** rate of these eight models:



Price-mileage curves, for eight selected car models

Model variant Price-Mileage ranking

- ▶ This graphic is not clear enough, every model has **multiple variants**
- ▶ To narrow down the **least depreciating** better models, we calculate the **price-mileage rate** for **every make-model-variant** of these eight models, and rank them



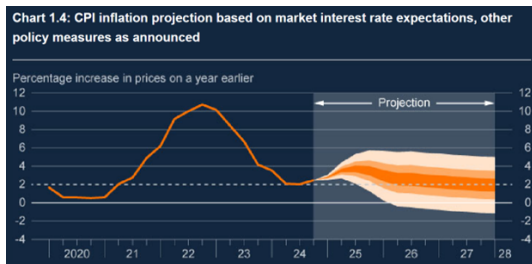
	car	avg_rate	rank
1	Audi_A4_TFSI	4.745493e+01	1
2	Vauxhall_Insignia_SRI	2.471312e+01	2
3	Volkswagen_Passat_R-Line	1.633762e+01	3
4	BMW_3_Series_330d	1.632473e+01	4
5	Audi_A4_Technik	1.141089e+01	5
6	Audi_A4_Sport Edition	1.043899e+01	6
7	Ford_Mondeo_EcoBlue	8.178241e+00	7
8	Volkswagen_Passat_Alltrack	7.050277e+00	8
9	Volkswagen_Passat_TSI	5.690001e+00	9
10	SKODA_Superb_TSI iV	5.101106e+00	10

Top 10 price-mileage rate ranking

- ▶ We will pick the **top 3** least depreciating model variants for analysis: Audi A4 TFSI, Vauxhall Insignia SRi, and Volkswagen Passat R-Line

Uncertainty and fan plots

- To visualise uncertainty, we can use a chart used by the Bank of England, the **fan plot**, which depends on mode, uncertainty, and skew, is modelled from an **asymmetrical normal distribution**, and we believe is **suitable** to represent the **uncertain** price distributions



an example of a *fan plot* used to represent CPI projections

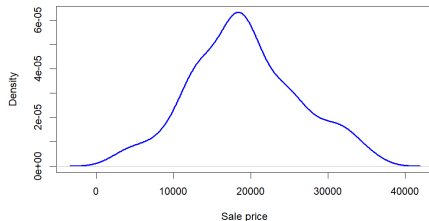
To achieve this, we will collect price density **statistics** (median, s.d., skew) of all model variants of a given age, and extend this to all other ages.

Audi A4 TFSI - density distributions

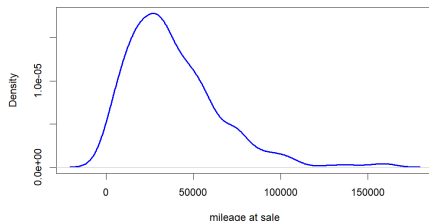
- From these results, we gather that, for this model variant, **sale prices** can go as high as £40,000, though the **most likely price** is just below £20,000. These cars are frequently sold at a **mileage** between 0 and 100,000, with the **most likely mileage** being about 30,000.



Density Distribution of a major retailer car prices, 2007 to 2022,
Audi A4 TFSI



Density Distribution of a major retailer car mileages, 2007 to 2022,
Audi A4 TFSI

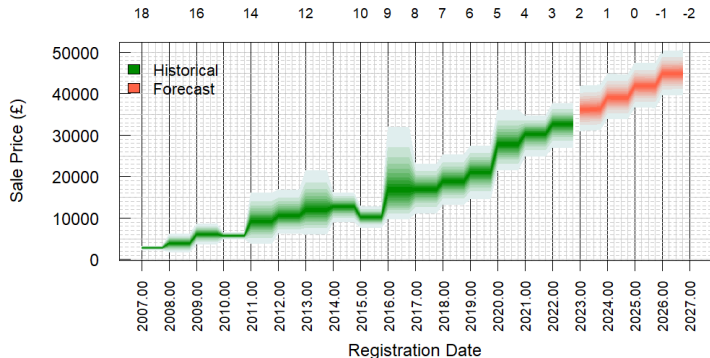


Price-Age distributions for Audi A4 TFSI ages

**Sale Price Distribution by Registration Date, for
a major retailer, plus SARIMA forecast, 2023-26**

Car: Audi A4 TFSI

Age (as of 2025)



	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
lower_bound	2392.306	1710.332	3738.294	4931.092	3856.779	6272.938	6136.663	9128.796	7777.535	9750.521	11136.94	13213.81	14668.22	21548.17	24891.78	27054.32
median	2820.000	3895.873	6087.221	5747.000	9190.845	10614.877	12031.930	12707.118	10245.000	16805.142	16954.44	18860.55	21018.26	27778.52	30269.64	32657.93
upper_bound	3247.694	6083.432	8660.269	6562.908	16106.271	16798.369	21434.556	16094.078	12712.465	31817.977	23013.18	25348.61	27422.04	36003.49	34674.80	37557.82

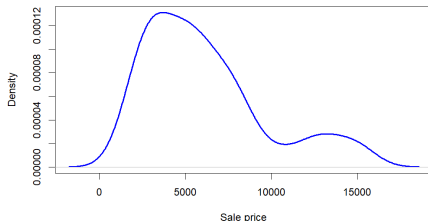
Sale price distribution for Audi A4 TFSI, every year, 2007 to 2022

Vauxhall Insignia SRi - density distributions

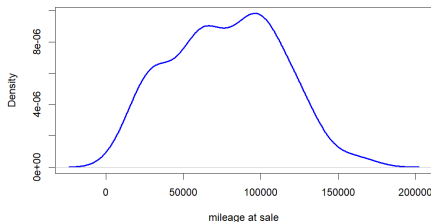
- From these results, we gather that, for this model variant, **sale prices** can go as high as £18,000, though the **most likely price** is just below £5,000. These cars are frequently sold at a **mileage** between 0 and 150,000, with the **most likely mileage** being about 100,000.



Density Distribution of a major retailer car prices, 2007 to 2022,
Vauxhall Insignia SRi



Density Distribution of a major retailer car mileages, 2007 to 2022,
Vauxhall Insignia SRi

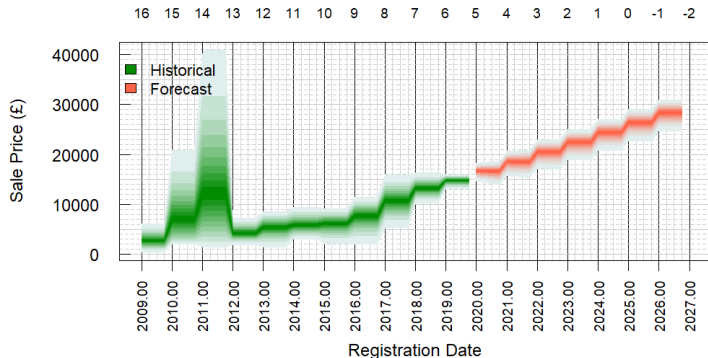


Price-Age distributions for Vauxhall Insignia SRi ages

**Sale Price Distribution by Registration Date, for
a major retailer, plus SARIMA forecast, 2020-26**

Car: Vauxhall Insignia SRi

Age (as of 2025)



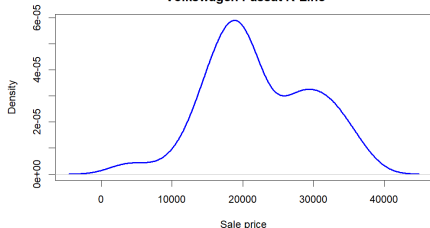
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
lower_bound	380.8326	2027.566	1482.907	1940.687	1440.554	3007.782	2118.797	2083.331	5338.312	10306.73	13153.21
median	2738.1074	7148.786	12180.630	4277.584	5404.640	5878.864	6146.794	7675.089	10682.804	13244.85	14792.49
upper_bound	6090.8363	20864.645	40831.705	7248.937	8598.492	9388.707	9166.377	11436.322	15878.975	16293.52	15975.27

Volkswagen Passat R-Line - density distributions

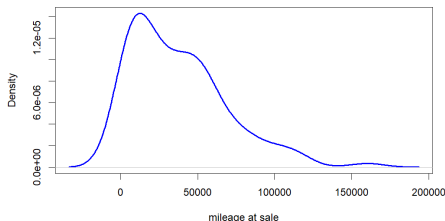
- From these results, we gather that, for this model variant, **sale prices** can go as high as £40,000, though the **most likely price** is just below £20,000. These cars are frequently sold at a **mileage** between 0 and 125,000, with the **most likely mileage** being about 20,000.



Density Distribution of a major retailer car prices, 2007 to 2022, Volkswagen Passat R-Line



Density Distribution of a major retailer car Mileages, 2007 to 2022, Volkswagen Passat R-Line

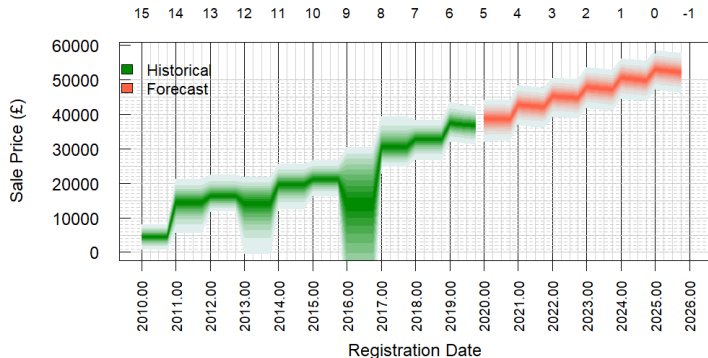


Price-Age distributions for Volkswagen Passat R-Line ages

**Sale Price Distribution by Registration Date, for
a major retailer, plus SARIMA forecast, 2020-26**

Car: Volkswagen Passat R-Line

Age (as of 2025)

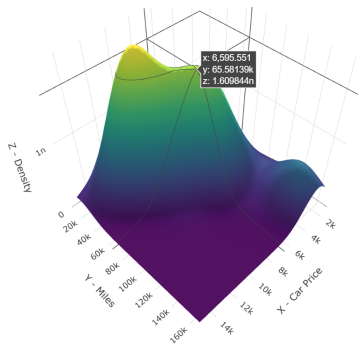


	2010	2012	2013	2014	2015	2016	2017	2018	2019
lower_bound	866.4721	5698.898	11913.28		12691.93	16384.74		24886.07	26941.01
median	4492.0000	14422.299	16359.70	14191.4176	19592.70	21235.64	13945.52	30503.45	32810.85
upper_bound	8117.5279	21173.056	22437.06	21891.6865	25605.51	26640.75	30339.67	39444.85	38267.34

Sale price distribution for Volkswagen Passat R-Line, every year, 2010 to 2019 (no 2011)

Our choice of car cost estimate

- ▶ We believe that a car model variant that has a **positively skewed** sale price distribution, and a **negatively skewed** mileage distribution, is our **best choice**, as this can mean **low price** and **long durability**
- ▶ Out of the three top cars we chose, the **Vauxhall Insignia SRi** is our best match, and we decide to go for this model variant
- ▶ A **3D density surface** of **sale prices** and **mileage** for this model variant hints that a **possible best deal** could be a car with **66,000 miles** and a **cost of £6,600**
- ▶ This mileage points to a **2013** car, the price distribution points to a price range of **£1440.55** to **£8598.49**, and averaging **£5404.64**.



Density of Vauxhall Insignia SRi cars by mileage and price

Transport cost quotes, vehicle and transit insurance

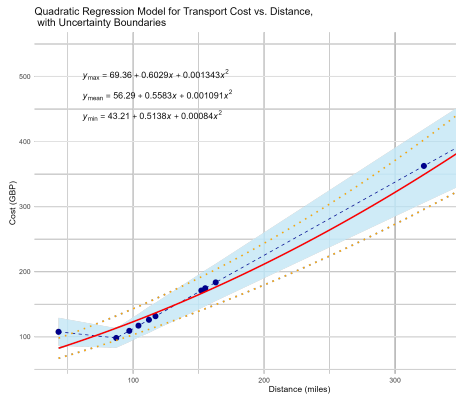
- ▶ We get in contact with a logistics provider, and they provide us with the following **prices per X miles** for their **transport** services:
- ▶ We **compare** these ranges with a few providers, and **assume** these estimates **represent** the current market
- ▶ We also require the car to be insured on the day of transportation. An insurer (such as the RAC) quotes **£35.07** to insure a car for **one day**. This figure will **vary** on **many factors**, so we take many **quote samples** and pick the most **comprehensive** option (we take £35.07 as average)
- ▶ We need a provider whose services **include transit insurance** (e.g., AA claims to cover up to 50k in losses), so we can assume **damage** or **loss** risk during transit to be covered

A car delivery quote from a provider

Distance (miles)	Min GBP	Max GBP	Mean GBP
43	86	129	107.5
87	83	113	98.0
97	92	126	109.0
104	99	135	117.0
112	106	146	126.0
117	111	152	131.5
152	144	198	171.0
155	147	202	174.5
163	155	212	183.5
322	306	419	362.5
411	390	534	462.0

Regression model for transport cost, sale distance rank

- ▶ From provider's estimates, we use **linear regression** of the **transportation cost distribution** to build a model to illustrate the uncertainty involved. We use this model to predict transportation costs later, when we know our target location.
- ▶ From our car data, we also work out distances from the mile **zero point** (Charing Cross) to locations of all Vauxhall Insignia SRi **seller locations**, to help us select a car that is **cheapest to transport** after purchase (see the next slide)



Regression model for transport cost vs. distance

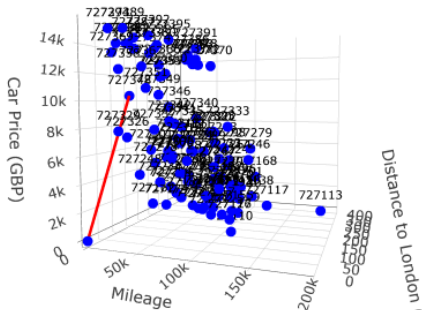
Our target car

- ▶ To decide which Vauxhall Insignia RSi to purchase from the car data, we can calculate the **euclidean distances** from the (0,0,0) origin of a price-mileage-distance subset, then **rank** all entries according to this euclidean distance. and choose a car with **low scores** as possible
- ▶ As an example, our target is a car in Watford, **15 miles** away, registered in **2014**, **28,549 miles** on the clock, and a sale price of **£7878**

X	car_price	miles	dist_to_london_miles	distance_from_origin	make	model	variant	car_seller_location	postcode	year
727348	10395	15205	98.806685	18418.95	Vauxhall	Insignia	SRi	Loughborough	LE11	2016
727384	14250	13067	60.311880	19334.23	Vauxhall	Insignia	SRi	Northampton	NN1	2018
727394	15199	12234	36.900189	19511.07	Vauxhall	Insignia	SRi	Leeds	ME17	2018
727369	13480	15847	36.900189	20804.79	Vauxhall	Insignia	SRi	Leeds	ME17	2017
727383	14200	17082	397.429668	22216.95	Vauxhall	Insignia	SRi	Aberdeen	AB10	2017
727395	15500	18600	260.558940	24213.18	Vauxhall	Insignia	SRi	Carlisle	CA1	2018
727379	14000	21548	344.332205	25698.93	Vauxhall	Insignia	SRi	Grangemouth	FK3	2018
727386	14500	21861	351.835111	26235.04	Vauxhall	Insignia	SRi	Alloa	FK10	2018
727374	13880	22291	154.863268	26259.61	Vauxhall	Insignia	SRi	Hull	HU1	2018
727387	14795	21836	93.805174	26376.35	Vauxhall	Insignia	SRi	Redditch	B96	2019
727381	14200	23607	60.311880	27548.76	Vauxhall	Insignia	SRi	Northampton	NN1	2018
727366	12999	26204	181.222821	29251.61	Vauxhall	Insignia	SRi	Burnley	BB10	2018
727382	14200	25727	60.311880	29385.75	Vauxhall	Insignia	SRi	Northampton	NN1	2018
727329	7878	28549	15.272325	29616.02	Vauxhall	Insignia	SRi	Watford	WD1	2014
727351	10999	31000	106.418042	32893.61	Vauxhall	Insignia	SRi	Bristol	BS1	2016
727358	12189	30627	10.715207	32963.39	Vauxhall	Insignia	SRi	Southall	UB1	2018

Car sale, plus transportation and insurance cost estimate

- ▶ We make use of the **price distribution** for the 2014 versions of these model variants, and find that the **price range** for a car of this profile could vary from £3007.78 and £9388.71
- ▶ The price range of this car indicates that availability of cars of similar age and price, above and below, is also **high** - losing this listing poses no major opportunity loss risk, there are plenty of similar offers
- ▶ We can estimate the **cost a car** of same model variant, age, 15 miles from London, plus **transport** and **insurance** costs (note, we do not have a model for insurance prices this time):
 - ▶ $(7878 + 64.91 + 35.07) = \text{£}7977.98$ (most likely scenario)
 - ▶ $(7878 + 64.91 + 35.07) + (1510.71 + 13.83) = \text{£}9502.52$ (worst case)
 - ▶ $(7878 + 64.91 + 35.07) - (4870.22 + 13.79) = \text{£}3121.55$ (best case)



Euclidian distance from (0,0,0) to the closest car available

Finance options

- ▶ With the car purchase estimate found, we now explore our **finance options**, and whether we can **afford** this car
- ▶ Now, we **compare** the different deals that **borrowing £8600** for 3, 4, or 5 years can offer. We assume **non-business banking** conditions

Finance Type	APR (%)	Deposit Needed (£)	Monthly Payments (£)	Term (Months)	Total Payable (£)	Own Car Straight Away	Own Car at End
Santander-5-years	6.00	0	165.63	60	9937.80	yes	yes
Santander-4-years	6.00	0	201.35	48	9664.80	yes	yes
Santander-3-years	6.00	0	261.01	36	9396.36	yes	yes
Barclays-5-years	6.50	0	167.52	60	10051.20	yes	yes
Barclays-4-years	6.50	0	203.21	48	9754.08	yes	yes
Barclays-3-years	6.50	0	262.85	36	9462.60	yes	yes
HSBC-5-years	6.40	0	167.14	60	10028.50	yes	yes
HSBC-4-years	6.40	0	202.84	48	9736.24	yes	yes
HSBC-3-years	6.40	0	262.49	36	9449.47	yes	yes
PCP-Carfinance247-5-years	11.90	1	149.69	60	12072.04	No	Optional
HP-Carfinance247-5-years	11.90	1	188.24	60	11305.40	No	Yes
PCP-Moneysupermarket-5-years	10.50	1	144.00	60	11597.00	No	Optional
HP-Moneysupermarket-5-years	9.90	1	180.00	60	10801.00	No	Yes

Finance Type	Secured (Against Car)	Excess Mileage Charges	Balloon/lump sum (£)	Fees (sum) (£)	Transport cost (once) (£)	Insurance cost (once) (£)
Santander-5-years	no	no	0.00	0	64.91	35.07
Santander-4-years	no	no	0.00	0	64.91	35.07
Santander-3-years	no	no	0.00	0	64.91	35.07
Barclays-5-years	no	no	0.00	0	64.91	35.07
Barclays-4-years	no	no	0.00	0	64.91	35.07
Barclays-3-years	no	no	0.00	0	64.91	35.07
HSBC-5-years	no	no	0.00	0	64.91	35.07
HSBC-4-years	no	no	0.00	0	64.91	35.07
HSBC-3-years	no	no	0.00	0	64.91	35.07
PCP-Carfinance247-5-years	Yes	Yes	3239.33	10	64.91	35.07
HP-Carfinance247-5-years	Yes	No	0.00	10	64.91	35.07
PCP-Moneysupermarket-5-years	Yes	Yes	3100.00	0	64.91	35.07
HP-Moneysupermarket-5-years	Yes	No	0.00	0	64.91	35.07

Monthly payments

To **reassure**, the monthly repayment amount M_i (for each i finance type), which concern us most, is calculated using the **loan amortisation** formula:

$$M_i = P_i \cdot \frac{r_i(1 + r_i)^{TM_i}}{(1 + r_i)^{TM_i} - 1}$$

- ▶ M_i = **Monthly Payment** for option i
- ▶ P_i = **Principal Amount** financed for option i (£8600)
(Note: we still borrow this amount, as the density of Vauxhall Insignia SRi cars by mileage and price indicate the price of 'best deal' car can go up to this amount, in case we need to find a different seller)
- ▶ r_i = **Interest Rate** for option i , calculated as $\frac{APR_i}{12 \times 100}$
(Note: we only consider fixed interest finance options)
- ▶ TM_i = **Term** (in months) for option i .

With various continuous and discrete variables to consider, we determine we can formulate this as a **Mixed Integer Program** problem (MIP). A good course of action is to run this problem through **Excel Solver**, which will use a form of the Simplex Algorithm to find an **optimal solution**.

MIP formulation

Set I of i Finance Options:

- ▶ Santander 5 years
- ▶ Santander 4 years
- ▶ Santander 3 years
- ▶ Barclays 5 years
- ▶ Barclays 4 years
- ▶ Barclays 3 years
- ▶ HSBC 5 years
- ▶ HSBC 4 years
- ▶ HSBC 3 years
- ▶ PCP Carfinance247 5 years
- ▶ HP Carfinance247 5 years
- ▶ PCP Moneysupermarket 5 years
- ▶ HP Moneysupermarket 5 years

Decision Variables

- ▶ $x_i \in \{0, 1\}$: Binary variable indicating whether finance option $i \in I$ is selected (1 if selected, 0 otherwise).

Parameters

- ▶ APR_i : Annual Percentage Rate for option i .
- ▶ D_i : Deposit Needed for option i .
- ▶ M_i : Monthly Payments for option i .
- ▶ TM_i : Term in Months for option i .
- ▶ C_i : Total Payable for option i .
- ▶ OB_i : Ownership (at start) indicator for option i (1 if yes, 0 otherwise).
- ▶ OA_i : Ownership (at end) indicator for option i (1 if yes, 0 otherwise).
- ▶ S_i : Secured indicator for option i (1 if secured against car, 0 otherwise).
- ▶ E_i : Excess Mileage Charges indicator for option i (1 if applies, 0 otherwise).
- ▶ B_i : Balloon/lump sum payment associated with option i .
- ▶ F_i : Fees (sum) for option i .
- ▶ T_i : Transportation costs for option i (one-off, fixed at £64.91 for all $i \in I$)
- ▶ IC_i : Insurance costs for option i (one-off, fixed at £35.07 for all $i \in I$)
- ▶ $M_{\max} = £500$: Maximum allowable monthly payment
- ▶ $TM_{\max} = £60$: Maximum allowable term in months

Objective Function

We want to **minimise** the total cost, which includes the total payable, any applicable balloon and fees, transportation costs, and insurance costs:

$$\min \sum_{i \in I} (C_i + B_i \cdot OA_i + T_i + IC_i + F_i) \cdot x_i \quad (1)$$

Constraints

- ▶ $\sum_{i \in I} x_i = 1$ (Select exactly one finance option)
- ▶ $\sum_{i \in I} M_i \cdot x_i \leq M_{\max}$ (Monthly budget constraint: £500)
- ▶ $\sum_{i \in I} TM_i \cdot x_i \leq TM_{\max}$ (Term constraint: ≤ 60 months)
- ▶ $\sum_{i \in I} (OA_i + OB_i) \cdot x_i \geq 1$ (Ownership requirement)
- ▶ $\sum_{i \in I} E_i \cdot x_i = 0$ (No excess mileage charges)
- ▶ $\sum_{i \in I} D_i \cdot x_i = 0$ (No deposit needed)
- ▶ $\sum_{i \in I} S_i \cdot x_i = 0$ (Not secured against car)
- ▶ $\sum_{i \in I} B_i \cdot x_i = 0$ (No balloon payment)
- ▶ $x_i \in \{0, 1\} \quad \forall i \in I$ (Binary variables)

MIP in Excel, formulation

- ▶ We **compile** a table in Excel, with all the data available to us
- ▶ Column A contains our **finance choices** for this purchase
- ▶ columns B to N contain all the **variables and conditions** of the deals
- ▶ Column O will display a '1' for the **optimal finance choice**
- ▶ columns P to U contains values **converted** from yes/no to 1/0
- ▶ column V contains formulae (not seen here) that **reference each constraint**, labelled in column W, of the formulation in cell notation

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Finance Type	APR (%)	Deposit	Monthly Pay	Term	Total Pay	Own Car	Own Car	Secured	Excess Mile	Balloon	Fees	Transport	Insurance	X (Decision)	Binary converters					Simplex formulae		
2	Santander-5-years	6	0	165.63	60	9937.8	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	OBJECTIVE
3	Santander-4-years	6	0	201.35	48	9664.8	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Select exactly one finance option
4	Santander-3-years	6	0	261.01	36	9396.36	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Monthly budget constraint
5	Barclays-5-years	6.5	0	167.52	60	10051.2	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Term constraint
6	Barclays-4-years	6.5	0	203.21	48	9754.08	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Ownership requirement
7	Barclays-3-years	6.5	0	262.85	36	9462.6	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Avoid excess mileage charges
8	HSBC-5-years	6.4	0	167.14	60	10028.5	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	No deposit needed
9	HSBC-4-years	6.4	0	202.84	48	9736.24	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	No secured against the car
10	HSBC-3-years	6.4	0	262.49	36	9449.47	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	No balloon payment
11	PCP-Carfinance247-5-years	11.9	1	149.69	60	12072	No	Optional	Yes	Yes	3239.3	10	64.91	35.07	0	0	1	1	1	1	1	1	
12	HP-Carfinance247-5-years	11.9	1	188.24	60	11305.4	No	Yes	Yes	No	0	10	64.91	35.07	0	0	1	1	0	0	1	1	
13	PCP-Moneysupermarket-5-years	10.5	1	144	60	11597	No	Optional	Yes	Yes	3100	0	64.91	35.07	0	0	1	1	1	1	1	1	
14	HP-Moneysupermarket-5-years	9.9	1	180	60	10801	No	Yes	Yes	No	0	0	64.91	35.07	0	0	1	1	0	0	1	1	

MIP in Excel, Solver

- Next, we open Solver and select the **decision variables**, the **objective function**, and the **constraints**

	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
	Deposit	Monthly Pay	Term	Total Pay	Own Car	Own Car	Secured	Excess Mile	Balloon/Fees	Transport	Insurance	x ₁ (Decision)	Binary converters					Simplex formulae				
2	0	165.63	60	9937.8	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	OBJECTIVE	
3	0	201.35	48	9664.8	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Select exactly one finance option	
4	0	261.01	36	9396.36	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Monthly budget constraint	
5	0	167.52	60	10051.2	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Term constraint	
6	0	203.21	48	9754.08	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Ownership requirement	
7	0	262.85	36	9462.6	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Avoid excess mileage charges	
8	0	167.14	60	10028.5	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	No deposit needed	
9	0	202.84	48	9736.24	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	Not secured against the car	
10	0	262.49	36	9449.47	yes	yes	no	no	0	0	64.91	35.07	0	1	1	0	0	0	0	0	No balloon payment	
11	1	149.69	60	12072	No	Optional	Yes	Yes	3239.3	10	64.91	35.07	0	0	1	1	1	1	1	1		
12	1	188.24	60	11305.4	No	Yes	Yes	No	0	10	64.91	35.07	0	0	1	1	0	0	1			
13	1	144	60	11597	No	Optional	Yes	Yes	3100	0	64.91	35.07	0	0	1	1	1	1	1			
14	1	180	60	10801	No	Yes	Yes	No	0	0	64.91	35.07	0	0	1	1	0	0	1			
15														Ow	Sec	Exc	Bal	Des				

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

\$O\$2:\$O\$14 = binary

\$V\$10 = 0

\$V\$3 = 1

\$V\$4 <= 500

\$V\$5 <= 60

\$V\$6 >= 1

\$V\$7 = 0

\$V\$8 = 0

\$V\$9 = 0

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: **Simplex LP**

Solution and report

- ▶ Solver indicates that our best finance choice is the **Santander 3 Years loan**, with 6% **APR**, no deposit, **monthly payments** of £261.01 over **36 months**, **£9396.36 total loan payable**, we own the car from the start, unsecured loan, no excess mileage, no balloon or extra fees, a one-off transport fee of £64.91, and a one-off insurance fee of £35.07
- ▶ Our final estimate for the purchase of a 2014 Vauxhall Insignia SRi, transported from Watford to central London, registered in 2014, with 28,549 miles on the clock, is **£9496.34**
- ▶ The Excel **Solver report** file, along with a **sensitivity analysis report** (not included here) can be handed to stakeholders, alongside a **formal document**, to inform and help make **better decisions**

Conclusion

- ▶ This method gains more accuracy with more **comprehensive**, **cleaner**, regularly **updated** data
- ▶ A thorough analysis of **insurance** and **transport** cost **distributions** would provide better estimates
- ▶ **Monte Carlo methods** remain a contender for future improvement of cost, risk, and uncertainty distributions of our method
- ▶ The car models and variants chosen for analysis remain a **personal choice**, although a more **objective method** to analyse all cars could be found
- ▶ Only finance options **known to us** were considered, and loan quotes vary for different lenders, borrowers, and multiple factors
- ▶ Methodology is partly **empirical**, subject to review and improvement

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